Received: 7 June 2022 | Revised: 13 September 2022 | Accepted: 12 October 2022

DOI: 10.1002/hed.27232

CLINICAL REVIEW

WILEY

Efficacy of optical coherence tomography in the diagnosing of oral cancerous lesion: systematic review and meta-analysis

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Funding information
National Research Foundation of Korea, Grant/Award Numbers: 2019M3A9H2032424, 2019M3E5D5064110, 2020R1I1A1A01051844

Abstract

Non-invasive diagnostic tools that facilitate visualization of potentially malignant oral lesions and cancers have been introduced. Oral lesions detected by optical coherence tomography (OCT) were compared to reference results based on histological findings. The diagnostic odds ratio (DOR), along with summary receiver operating characteristic curve (SROC), area under SROC, sensitivity, specificity, and negative predictive values, were the outcomes. The DOR of OCT was 86.9190 (95% confidence interval [CI]: 38.7435, 194.9985), and the area under SROC was 0.951. OCT showed good sensitivity (0.9138; 95% CI: 0.8758, 0.9409) and specificity (0.9110; 95% CI: 0.8568, 0.9460), and a high negative predictive value (0.9225; 95% CI: 0.8863, 0.9478). Diagnostic sensitivity was higher when using artificial intelligence and automated algorithms compared to diagnoses made by clinicians. OCT is non-invasive, provides rapid results without radiation exposure, and can aid in the diagnosis and follow-up of oral cancer and oral precancerous lesions.

KEYWORDS

artificial intelligence, diagnosis, mouth neoplasms, optical coherence, precancerous conditions, tomography

1 | INTRODUCTION

Oral cancer can occur on the tongue, floor of the mouth, gingiva, gums, palate, and lip. There are approximately 350 000 cases worldwide every year, resulting in 170 000 deaths.[1] More than 90% of oral cancers arise from the squamous cells that line the inside of the mouth. The progression of oral potentially malignant lesions (OPMLs) to oral squamous cell carcinoma has been well documented.[2] OPML is often difficult to diagnose because it has a similar appearance to oral reactive and inflammatory tumors, and is often asymptomatic.[3] However, early detection of OPML is important for a good prognosis; therefore, clinicians should be able to differentiate between these lesions.[4] Although biopsy of the lesion is the most accurate diagnostic method, it is invasive and associated with morbidity.[5] Also, it is difficult to perform a biopsy of the lesion at every follow-up visit. Therefore, several non-invasive diagnostic tools that facilitate the visualization of OPML and oral cancer have been introduced.[3,6]

Optical coherence tomography (OCT) was introduced in 1991[7] and allows for real-time, non-invasive, label-free body imaging.[8] Clinically, it was initially used for

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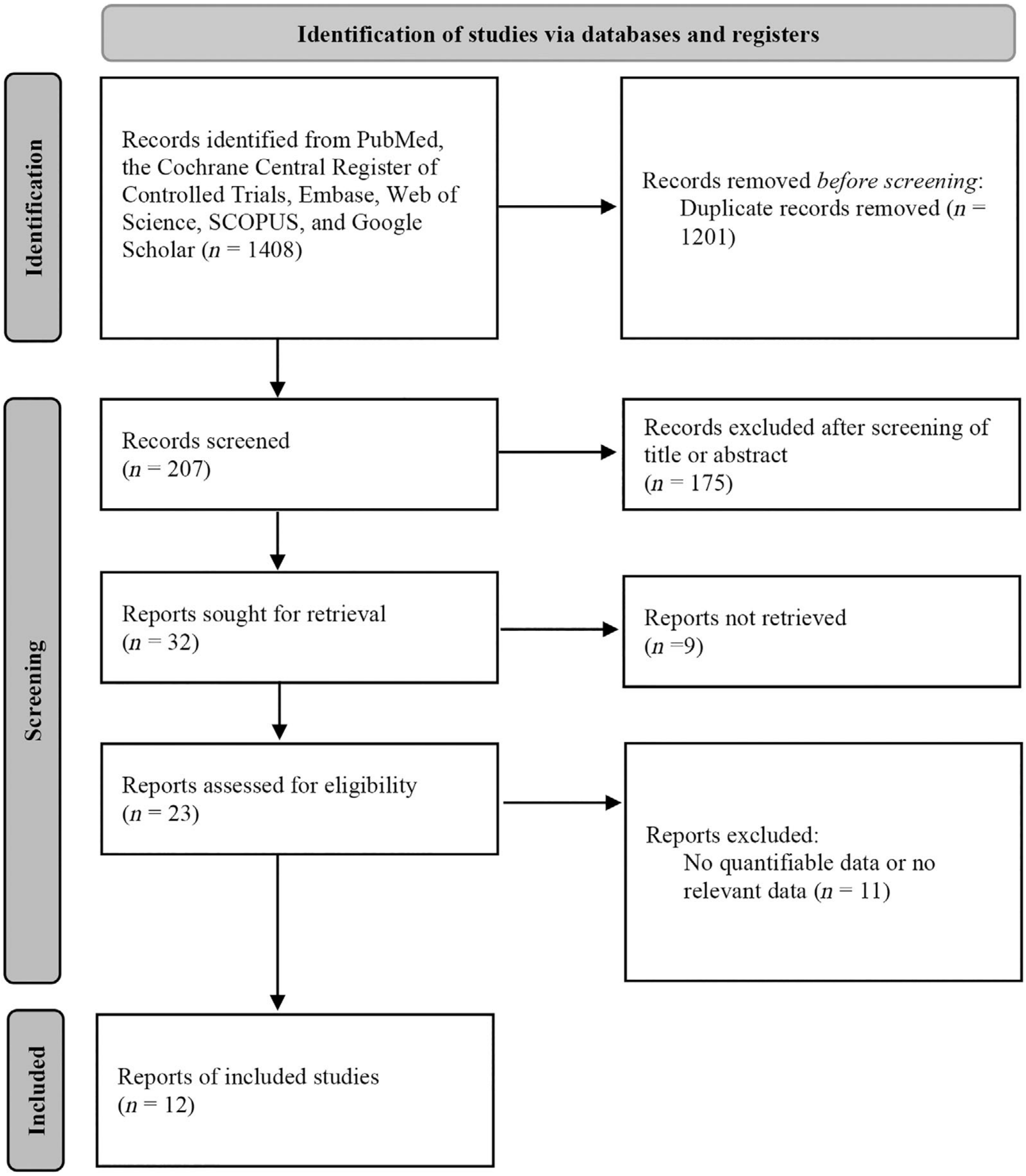

FIGURE 1 Flow diagram of the search strategy

glaucoma diagnosis,[9] but was soon applied to make diagnoses in other parts of the body.[10,11] OCT is used to detect early epithelial changes in oropharyngeal and laryngeal regions, because it provides high-resolution images of the tissue microanatomy and cellular structures of the mucosa.[12] In addition, automated algorithms[12,13] and artificial intelligence (AI)[14] have begun to be applied to oral cancer diagnosis.[8] Therefore, in this study, the diagnostic performance of OCT for oral cancer and precancerous lesions was investigated. We also investigated bivariate meta-analyses including subgroup analyses on the accuracy to detect high-risk patients according to diagnostic criteria (including oral cancer only or OPML) and judgment criteria (clinician, automatic diagnosis device, or AI-based diagnosis).

2 | MATERIALS AND METHODS

2.1 | Search strategy

The study was prospectively registered at Open Science Framework (https://osf.io/m4u5s/). The PubMed, Cochrane library, Web of Science, SCOPUS, Google Scholar, and Embase databases were searched up to May 2022. The search terms were as follows: "optical coherence tomography," "dysplasia," "oral precancer," "oral cancer," and "oral carcinoma." The titles and abstracts of candidate studies were reviewed by two independent reviewers (SHH and DHK). Studies not focusing on OCT for oral cancer were excluded. Differences of opinion between the two reviewers regarding study inclusion were resolved through discussion with a third reviewer (SWK).

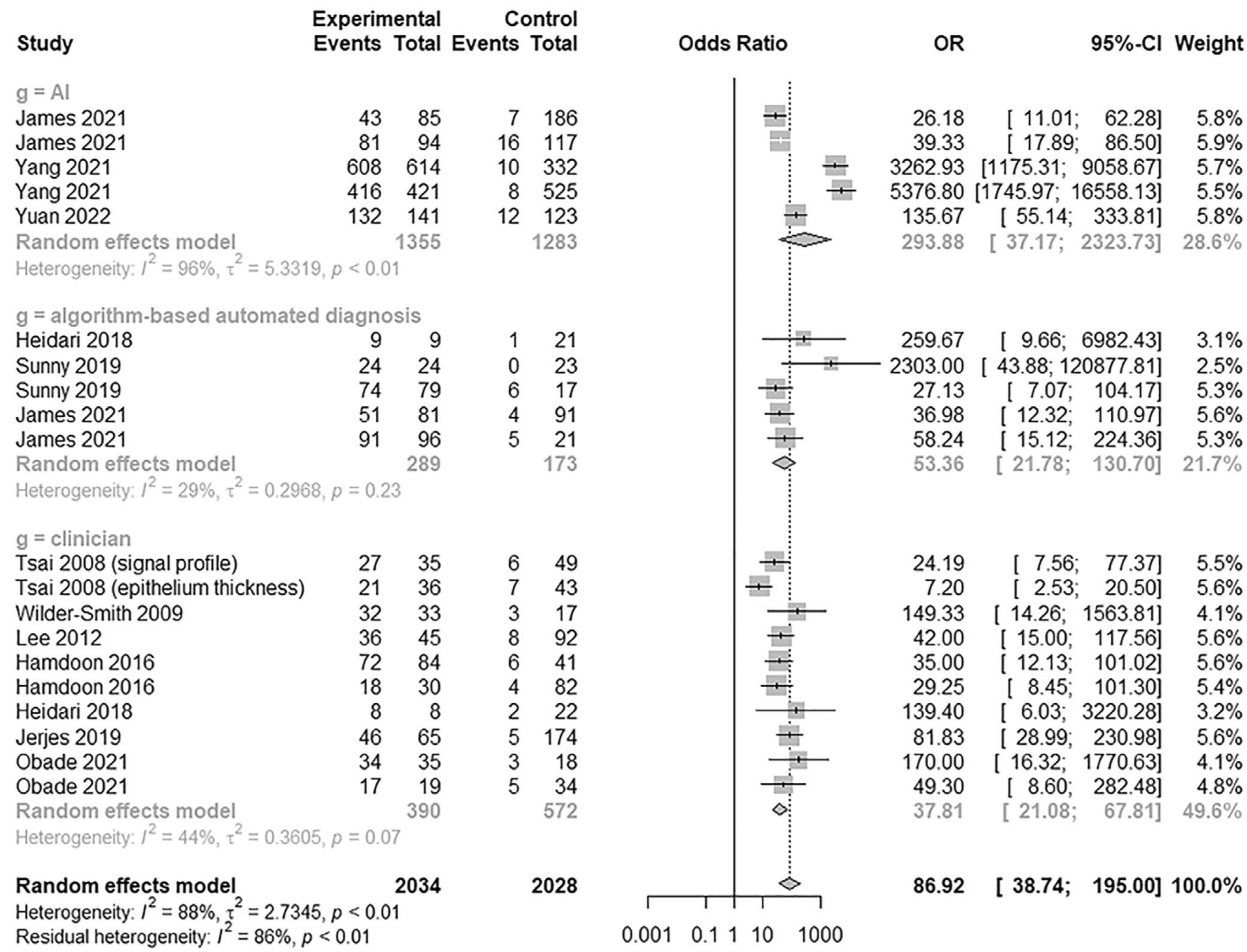

FIGURE 2 Forest plot of the diagnostic odds ratios of the included studies

2.2 | Study selection

Studies were included if they met the following criteria: inclusion of patients with clinically suspected oral tumors or lesions; use of OCT for diagnosis; comparison of OCT and histological (reference) findings; and availability of sensitivity and specificity data. Case reports, review articles, studies of other head and neck tumors, such as laryngeal or nasal tumors, and studies with a lack of relevant data were excluded. A flowchart of the search process is presented in Figure 1.

2.3 | Data collection and quality assessment

Data were extracted and recorded using a standard format.[15,16] All included studies collected demographic, true-positive, true-negative, false-positive, and false-negative data, and provided diagnostic odds ratios (DORs), summary receiver operating characteristic (SROC) curves, and area under the SROC curve (AUC) values.[8,12–14,17–23] The diagnostic power of OCT was compared to that of histology, as stated above. Random effects models considering inter- and intra-study reproducibility were used for the analysis.

DOR was calculated as (true-positives/false-positives)/(false-negatives/true-negatives); 95% confidence intervals (CIs) were also provided. A higher DOR value (range: 0 to infinity) indicates higher diagnostic accuracy; a value of 0–1 indicates an inverse correlation, and a value of 1 indicates that the performance of the diagnostic tool cannot be determined. We calculated the logarithm of each DOR to obtain an approximately normal distribution.[24] SROC curve analysis is the most commonly used method to calculate sensitivity and specificity when performing meta-analyses. With higher diagnostic accuracy, the SROC curve is located more toward the upper-left corner of the graph (i.e., the point where both sensitivity and specificity are 100%). AUC values range between 0 and 1, with higher values indicating better diagnostic performance. Study quality was assessed by the Quality Assessment of Diagnostic Accuracy Studies tool version 2 (QUADAS-2).

2.4 | Statistical analysis

R software (R Foundation for Statistical Computing, Vienna, Austria) was used for the meta-analysis. For inter-rater agreement assessment, reliability coefficients were transformed into Fisher z-values to normalize the distribution and achieve homogeneity of variance. Homogeneity was determined by the Q statistic. We generated forest plots of the sensitivity, specificity, and negative predictive values, and drew SROC curves.

3 | RESULTS

Twelve articles including 481 participants were included in the meta-analysis. The study characteristics are listed in Table S1, and the bias assessment results are presented in Table S2. Begg's funnel plots (Figure S1) indicated that there was no bias in the included studies. Egger's test also indicated that the likelihood of publication bias was low ($p > 0.05$).

3.1 | Diagnostic accuracy of OCT

The DOR of OCT was 86.9190 (95% CI: 38.7435, 194.9985; $I^2 = 87.7\%$) (Figure 2). The AUC was 0.951. OCT exhibited good sensitivity (0.9138; 95% CI: 0.8758, 0.9409; $I^2 = 73.6\%$) and specificity (0.9110; 95% CI: 0.8568, 0.9460; $I^2 = 89.6\%$), and a high negative predictive value (0.9225; 95% CI: 0.8863, 0.9478; $I^2 = 77.2\%$) (Figure 3). The area under the SROC curve value was 0.90–1.00, which indicates excellent diagnostic accuracy (Figure 4).[25] The correlation between sensitivity and the false-positive rate was 0.133, indicating no heterogeneity.

A subgroup analysis was conducted to compare the diagnostic power of OCT for high-risk lesions, including OPMLs (any dysplasia, carcinoma in situ, or invasive cancer), with that for oral cancer. For OCT-based screening for oral cancer, only the negative predictive value was significantly higher (0.9548 vs. 0.8848; $p = 0.0068$). OCT-based screening for oral cancer tended to be more sensitive (0.9316 vs. 0.8960; $p = 0.2509$) and specific (0.9257 vs. 0.8962; $p = 0.5007$), and showed superior diagnostic accuracy (139.2463 vs. 60.0535; $p = 0.3214$), than screening for OPMLs (Table 1).

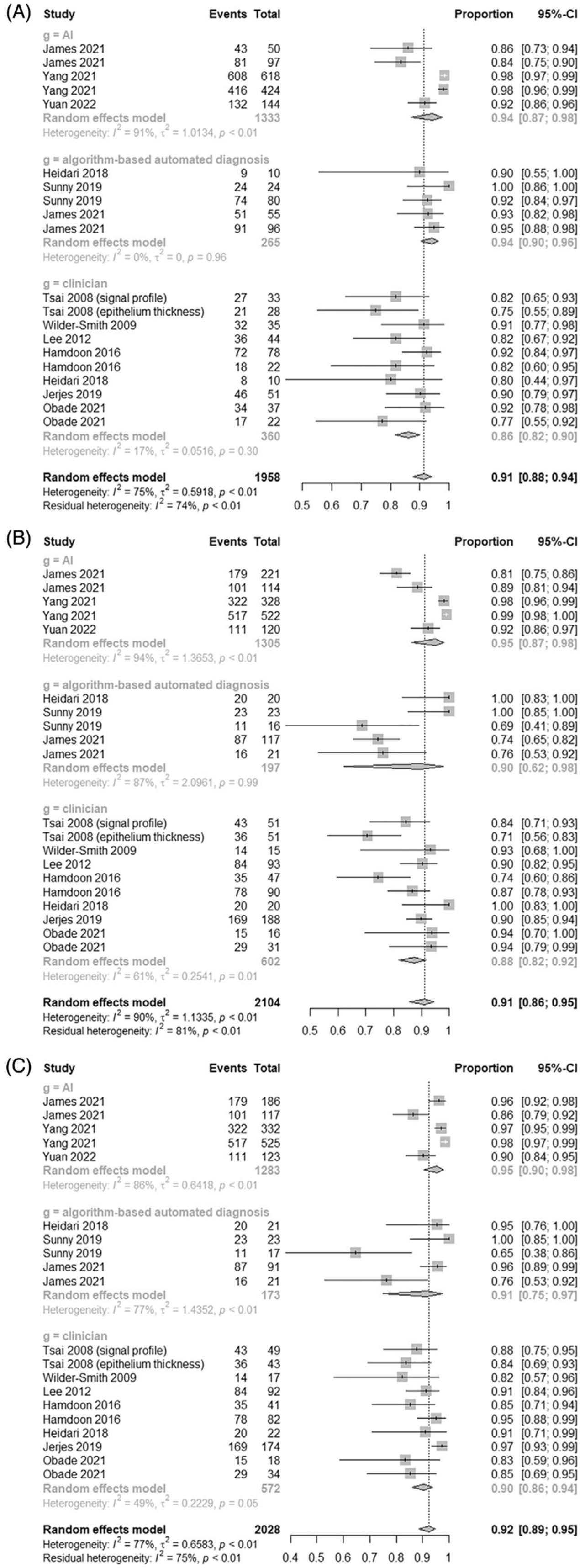

FIGURE 3 Forest plots of the sensitivity (A), specificity (B), and negative predictive values (C) of the included studies

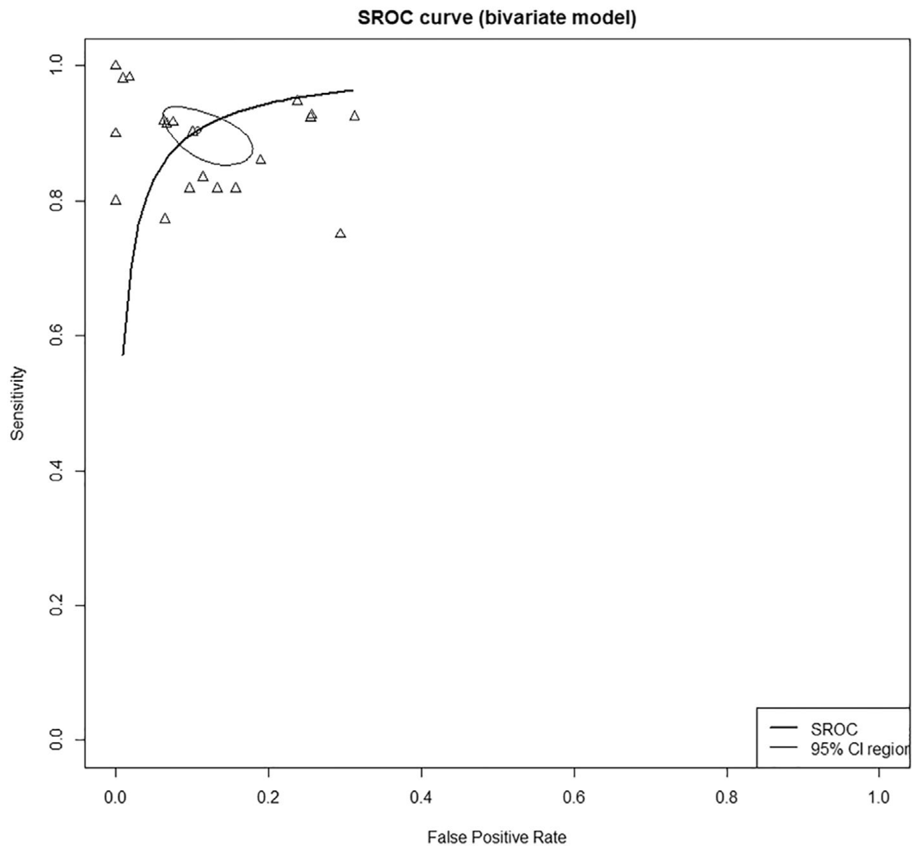

FIGURE 4 Area under the summary receiver operating characteristic of the included studies

Subgroup analysis was performed to compare the accuracy of clinician, automated algorithm, and AI judgments. Sensitivity was significant lower for clinician judgments than automated algorithm and artificial intelligence judgments of oral precancerous and cancerous lesions (0.8623 vs. 0.9396 and 0.9440; $p = 0.0050$). Moreover, specificity showed a trend toward being lower for clinician judgments (0.8759 vs. 0.8985 and 0.9493; $p = 0.2546$), as was the negative predictive value (0.9045 vs. 0.9148 and 0.9532; $p = 0.2287$) and diagnostic accuracy (37.8053 vs. 53.3577 and 293.8838; $p = 0.1631$) (Table 1).

4 | DISCUSSION

OCT is a non-invasive, high-resolution imaging modality that can reveal near-surface anomalies, including in epithelial and sub-epithelial tissues.[24] Both OCT and ultrasound provide real-time structural images. However, in contrast to ultrasound, OCT is based on a low-coherence interferometer, which is a non-contact optical sensing technology, and provides high-resolution images of subsurface tissue. It is also possible to obtain macroscopic epithelial and subepithelial images in real time.[18]

Clinical OCT images of human teeth and oral mucosa were first presented in 1998.[26,27] In 2004, OCT analysis of ex vivo images of a hamster buccal mucosa pouch with malignant mucosa was performed to assess the feasibility of using OCT scans to diagnose oral diseases.[25,28] The clinical applicability of OCT was confirmed by comparative analysis of human oral mucosa.[17,29] Since then, numerous investigations of the diagnostic accuracy of OCT for distinguishing oral cancerous lesions have been performed. Review articles have also summarized studies showing the efficacy of OCT for differentiating malignant tumors from benign and inflammatory mucosal lesions.[30,31] However, since no meta-analysis has analyzed the diagnostic power of OCT for OPMLs or oral cancer lesions, we performed the present bivariate meta-analysis. In addition, qualitative and quantitative OCT data may vary depending on the diagnostic category, and results may differ depending on whether the judgment is made by a trained clinician, automated algorithm, or AI. We performed a subgroup analysis to address this issue.

In this study, OCT showed a pooled sensitivity of 91%, pooled specificity of 91%, and pooled negative predictive value of 92%. Also, the AUC was 0.951, indicating excellent diagnostic accuracy.[25] This high diagnostic accuracy can be explained by the mechanism of OCT,

TABLE 1 Results of subgroup analysis of the diagnostic power of AI, automated algorithms and clinicians for high-risk lesions and oral cancer

Subgroup	No. of Study	DOR (95% CIs)	Sensitivity (95% CIs)	Specificity (95% CIs)	NPV (95% CIs)	AUC
	12	86.9190 (38.7435, 194.9985); $I^2 = 87.7\%$	0.9138 (0.8758, 0.9409); $I^2 = 73.6\%$	0.9110 (0.8568, 0.9460); $I^2 = 89.6\%$	0.9225 (0.8863, 0.9478); $I^2 = 77.2\%$	0.951
Subject of judgment						
Artificial intelligence	5	293.8838 [37.1677, 2323.7304]; $I^2 = 96.0\%$	0.9440 [0.8691, 0.9771]; $I^2 = 17.3\%$	0.9493 [0.8655, 0.9820]; $I^2 = 93.6\%$	0.9532 [0.9050, 0.9775]; $I^2 = 86.1\%$	
Automated device	5	53.3577 [21.7823, 130.7048]; $I^2 = 29.4\%$	0.9396 [0.9037, 0.9627]; $I^2 = 0.0\%$	0.8985 [0.6220, 0.9794]; $I^2 = 87.5\%$	0.9148 [0.7487, 0.9748]; $I^2 = 77.2\%$	
Inspector	10	37.8053 [21.0763, 67.8128]; $I^2 = 43.9\%$	0.8623 [0.8167, 0.8979]; $I^2 = 90.5\%$	0.8759 [0.8184, 0.9170]; $I^2 = 61.0\%$	0.9045 [0.8601, 0.9359]; $I^2 = 48.7\%$	
p		0.1631	0.005	0.2546	0.2287	
Application range						
Including oral pre-malignant lesions	11	60.0535 [20.1780, 178.7300]; $I^2 = 87.8\%$	0.8960 [0.8291, 0.9386]; $I^2 = 78.8\%$	0.8962 [0.8100, 0.9459]; $I^2 = 85.1\%$	0.8848 [0.8289, 0.9241]; $I^2 = 67.0\%$	
Only oral cancer	9	139.2463 [39.7058, 488.3302]; $I^2 = 88.2\%$	0.9316 [0.8888, 0.9587]; $I^2 = 63.0\%$	0.9257 [0.8482, 0.9653]; $I^2 = 91.7\%$	0.9548 [0.9228, 0.9739]; $I^2 = 67.3\%$	
p		0.3214	0.2509	0.5007	0.0068	

Abbreviations: AUC, area under the curve; CI, confidence interval; DOR, diagnostic odds ratio; NPV, negative predictive value.

which enables imaging of epithelial and subepithelial structures for analysis of epithelial thickness, margins, and macroscopic histopathological appearance. Since the oral mucosa is normally 0.2–1-mm thick, OCT with a tissue imaging depth of 1–3 mm is suitable for evaluating it.[28] Studies have suggested that the disruption of oral mucosal structures that occurs with progression from normal to invasive cancer is discriminable by OCT.[13] Also, as the tissue changes from normal to premalignant, the squamous epithelium tends to thicken.[29] OCT can identify epithelial stratification, as well as structural and thickness changes through high-resolution epithelial and subepithelial images; our results are consistent with recent studies.[21,31] OCT may be useful in clinical practice to identify the optimal biopsy site, follow up lesions, and identify high-risk groups.[28]

Several recent studies have reported automated image processing tools for classifying OCT images as normal, dysplastic, or malignant.[32] Using image processing methods, other studies analyzed the progression of dysplasia/carcinoma in in situ/invasive cancer based on epithelial thickness, the standard deviation of the scattering intensity signal of the epithelial lamina propria junction, and the cellular heterogeneity of the epithelium and basement membrane.[17,20,33] AI algorithms, including machine learning and deep learning algorithms, are the most commonly used image analysis techniques; they are being increasingly applied in the medical field. Models can learn through OCT images, and sophisticated modeling can be performed as data accumulate.[8] AI algorithms usually set diagnostic criteria through the following process. First, OCT images are acquired, and the images are normalized so that they can be trained and validated in the same environment (image processing). The images are divided into a model training set and a cross-validation set for training and validation. ROC curve analysis (representing the relationship between true and false positive rates) was performed on the cross-validation data set to find the optimal cut-off score for classification. The final diagnosis of the subject is determined according to the cut-off values. In our study, automated algorithms and AI were more diagnostically sensitive significantly compared to clinician judgments, and tended to have higher diagnostic power.

Extensive training is required before human observers can interpret OCT images.[34] Recent OCT systems can generate more than 40 images, and it is difficult to interpret such large volumes of clinical data.[32] When large numbers of images must be analyzed, the concentration of the observer may decline, and important images may be missed. In contrast, automated analytical approaches can handle large OCT data sets without fatigue (or the associated mistakes).[32] In addition, AI can provide quantitative data to aid clinical decision-making, by transforming diagnostic images with unclear margins. With sufficient data sets and algorithms, the diagnostic accuracy of AI could be similar, or even exceed, that of clinical experts.[31]

Oral precancerous lesions (dysplasia and carcinoma in situ) are significantly more likely to progress to invasive squamous cell carcinoma.[35] Therefore, screening tools are important for early detection. In our subgroup analysis, screening for OPML tended to be less sensitive (0.8960 vs. 0.9316) and specific (0.8962 vs. 0.9316) than that for oral cancer. Nevertheless, the generally high sensitivity and specificity indicate the potential utility of OCT for cancer screening and follow-up. However, since biopsies are readily available in oral lesions, OCT has not yet been widely used in these lesions. On the other hand, in the field of ophthalmology, where biopsy is difficult, OCT is being actively used in various diseases. However, in recent years, the trend is to use non-invasive tools for diagnosis and follow-up of diseases, and the diagnosis accuracy of OCT is improving. Therefore, in the future, OCT could be more clinically used for oral lesions. In order to expand clinical application, several additional studies are needed. First, it is necessary to standardize instrument settings and standardize image measurement methods for universal use of data. Also, additional research is needed to establish criteria for classification.

This study had several limitations. First, although several parameters were used to distinguish benign lesions, OPMLs, and malignant lesions, such as disruption of the epithelial layer, epithelial thickening, micro-invasion, heterogeneity of cell distribution, and disorder of the basement membrane,[17,18,20] standardized diagnostic parameters should be established to expand the clinical applications of diagnostic algorithms.[31] Second, several studies scanned ex vivo oral tissue from tumor sections or biopsies with an OCT system, and compared diagnoses based on OCT and histopathology. However, ex vivo studies do not account for potential movement artifacts or anatomical restrictions on the use of devices in the clinic.[18] Moreover, results differ between ex vivo and in vivo clinical examinations. Therefore, in vivo studies are needed to verify our results. Third, mechanical compression of the mucosa by the OCT probe may have affect the results, by decreasing tissue thickness or increasing contrast between layers, for example.[31,36]

5 | CONCLUSION

This meta-analysis demonstrated the high specificity and sensitivity of OCT, thus demonstrating its utility for early detection and diagnosis of cancerous oral lesions, and for follow-up of suspicious oral lesions and rapid screening of high-risk groups.

FUNDING INFORMATION

This research was supported by the Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Education (2020R1I1A1A01051844), and the Bio & Medical Technology Development Program of the National Research Foundation (NRF) funded by the Ministry of Science & ICT (2019M3A9H2032424, 2019M3E5D5064110). The sponsors had no role in the study design, data collection and analysis, decision to publish, or preparation of the manuscript.

CONFLICT OF INTEREST

No potential conflict of interest relevant to this article was reported.

DATA AVAILABILITY STATEMENT

The raw data of individual articles used in this meta-analysis are included in the main text or supplementary data.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Kim DH, Kim SW, Hwang SH. Efficacy of optical coherence tomography in the diagnosing of oral cancerous lesion: systematic review and meta-analysis. *Head & Neck*. 2023;45(2):473-481. doi:10.1002/hed.27232